

STATISTICS MASTER'S COMPREHENSIVE EXAMINATION

March 28th, 2020

Instructions: Attempt any five out of the following six questions on the 2 pages.
Mention which five you would like to have graded. Open book and notes.

Problem 1. Let $X_1, \dots, X_n$ be a random i.i.d. sample from an arbitrary discrete distribution P on $\{1, 2, 3\}$. Find a two-dimensional sufficient statistic.

Problem 2. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. samples with $X_i \sim N(0, 1)$ and $Y_i|X_i = x \sim N(x\theta, 1)$.

(1) Find the MLE $\hat{\theta}_n$ of θ .

(2) Find the limiting distribution of $\sqrt{n}(\hat{\theta}_n - \theta)$.

3. a. The effectiveness of a COVID-19 vaccine is being studied in a high risk area; of 90 people examined 18 had COVID-19 and 45 had been vaccinated. Of the 18 people with COVID-19 only 3 had been vaccinated. Can we conclude that vaccination and contracting COVID-19 are independent?
- b. A researcher studying the COVID-19 virus predicts that there are 3 different clades (A, B and C) and that they should exist in a 6:1:1 ratio. (NOTE - Each infected person only has one Clade of the virus) To test this assumption, 48 persons infected with COVID-19 are randomly sampled and the viruses tested with the following numbers observed: Clade-A $N=30$, Clade-B $N=6$, Clade-C $N=12$. Do these data suggest that the 6:1:1 ratio of Clade-A to Clade-B to Clade-C is reasonable?
- c. Use the data in part b to create a 95% confidence interval for the proportion of infected persons that have Clade-C.
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4. let X be a worker's test score from the FDA training class and Y be the standardized hourly output on N95 masks from that worker. Evaluation of $N=10$ independent workers gives the following:

$$\Sigma X = 550 \quad \Sigma Y = 680 \quad \Sigma X^2 = 38,500 \quad \Sigma Y^2 = 56,000 \quad \Sigma XY = 45,500$$

Assume a multivariate normal distribution holds.

- a. Determine the parameter estimates for the standard linear regression of Y on X
- b. Is there sufficient evidence to conclude that the slope of the above line is different from 0 at $\alpha=0.02$?
- c. Find the 98% confidence interval for the value of Y of the individual with $X = 40$
- d. What is the correlation between X and Y
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Problem 5

At a chemical plant that galvanizes electricity towers we measure two variables hourly, the concentration of a certain chemical and the loss of stack. We are concerned that these variables may contain outliers. The data is.

CONCENTRATION	LOSS
62	18
62	19
62	18
79	15
58	14
50	7
50	8
58	14
50	8
50	9
58	13
58	11
50	8
62	20
56	15
58	15
58	12

- (1) Use Grubbs' test and call x_j an outlier if $|z_j| > 3.5$, where z_j is the z-score corresponding to x_j . This is done separately for each variable in the data set.
- (2) Use the resistant z-score rule and call x_j an outlier if $|z_j^*| > 3.5$, where z_j^* is the resistant z-score corresponding to x_j . (Hint: The resistant z-score is similar to the z-score but instead of mean and standard deviation it uses the median and the median absolute deviation.)
- (3) Identify the outliers in (1) and (2) and explain why these two methods may produce different results. Draw a scatter plot of CONCENTRATION vs LOSS and search for outliers visually. Check if you get the same answer as the Grubbs' test or the resistant z-score rule.

Problem 6. Given a random sample of n independent observations $\{X_1, \dots, X_n\}$ from a distribution with density

$$f(x) = \begin{cases} \lambda k x^{k-1} e^{-\lambda x^k} & 0 < x < \infty \\ 0 & \text{Otherwise} \end{cases}$$

- (1) Write down the likelihood function for λ and the corresponding likelihood equations for estimating λ . Give the explicit form of the MLE of λ .
- (2) Write down the likelihood function for k and the corresponding likelihood equations for estimating k . Is there an explicit form for the MLE of k ?